

Some history

Quantifying the infectivity of human immunodeficiency virus

(viral infectivity assay/human immunodeficiency virus kinetics/gp120 blocking protein)

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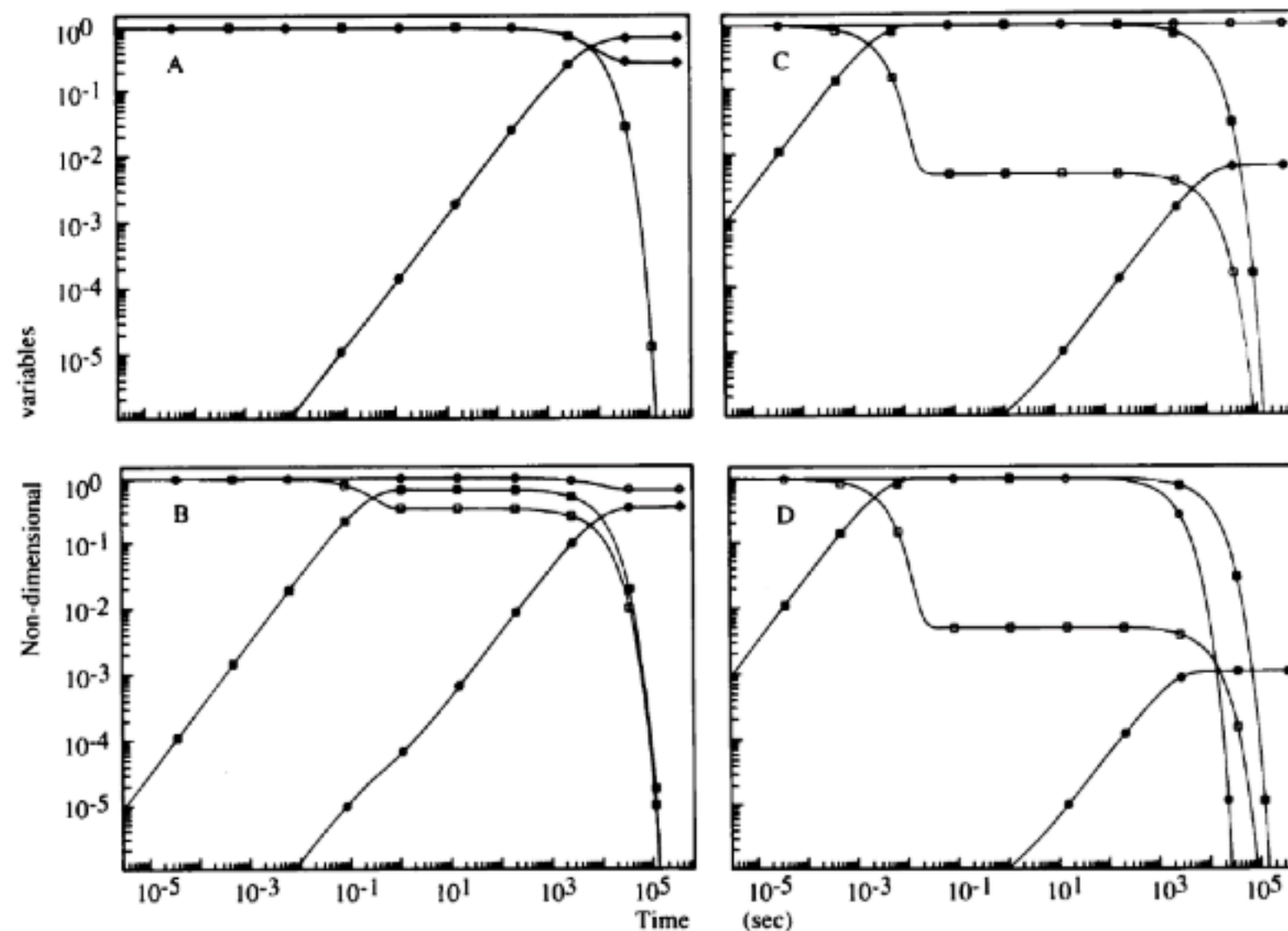
Communicated by Stirling A. Colgate, March 8, 1989 (received for review January 9, 1989)

$$\frac{di}{dt} = N\lambda f v,$$

$$\frac{dv}{dt} = -[N\lambda f + 1 - \sigma]v,$$

$$\frac{df}{dt} = -\gamma(\beta f - c) - \sigma f - \lambda f(1 - f),$$

$$\frac{dc}{dt} = \gamma(\beta f - c) - \sigma c + \lambda f c.$$



Dynamics of HIV Infection of CD4⁺ T cells

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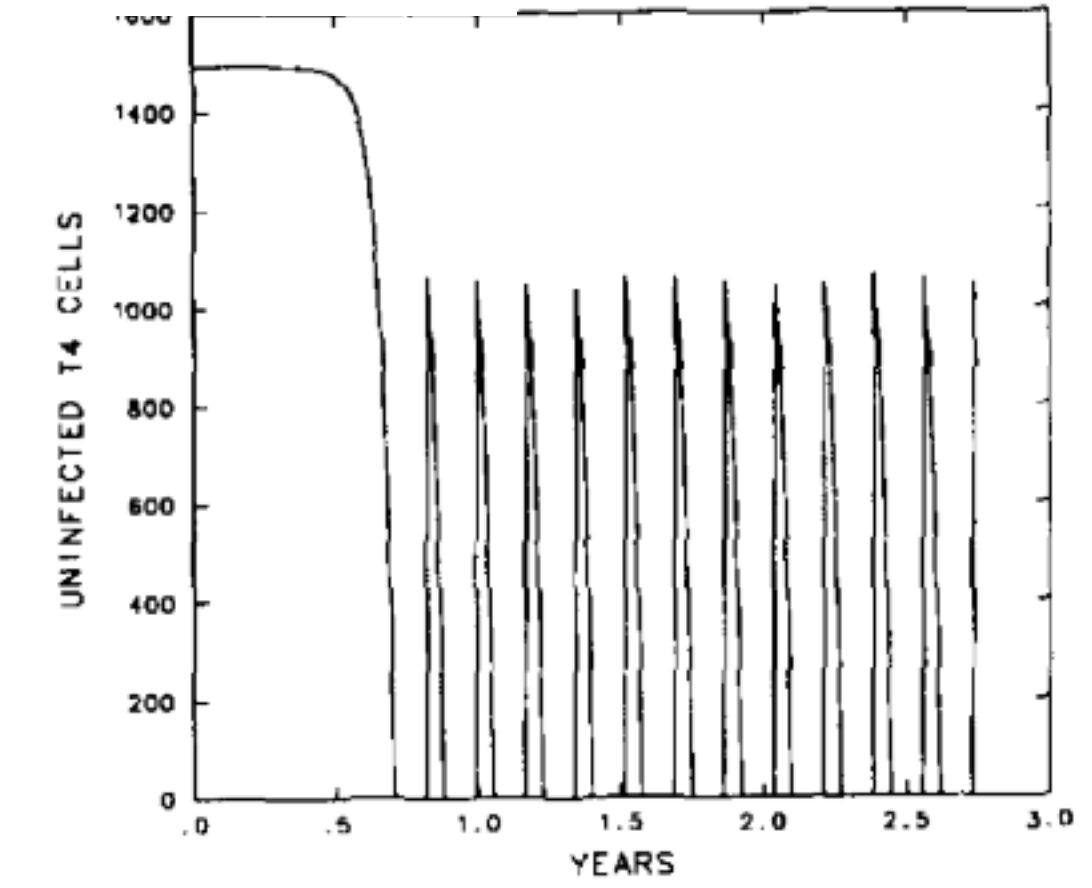
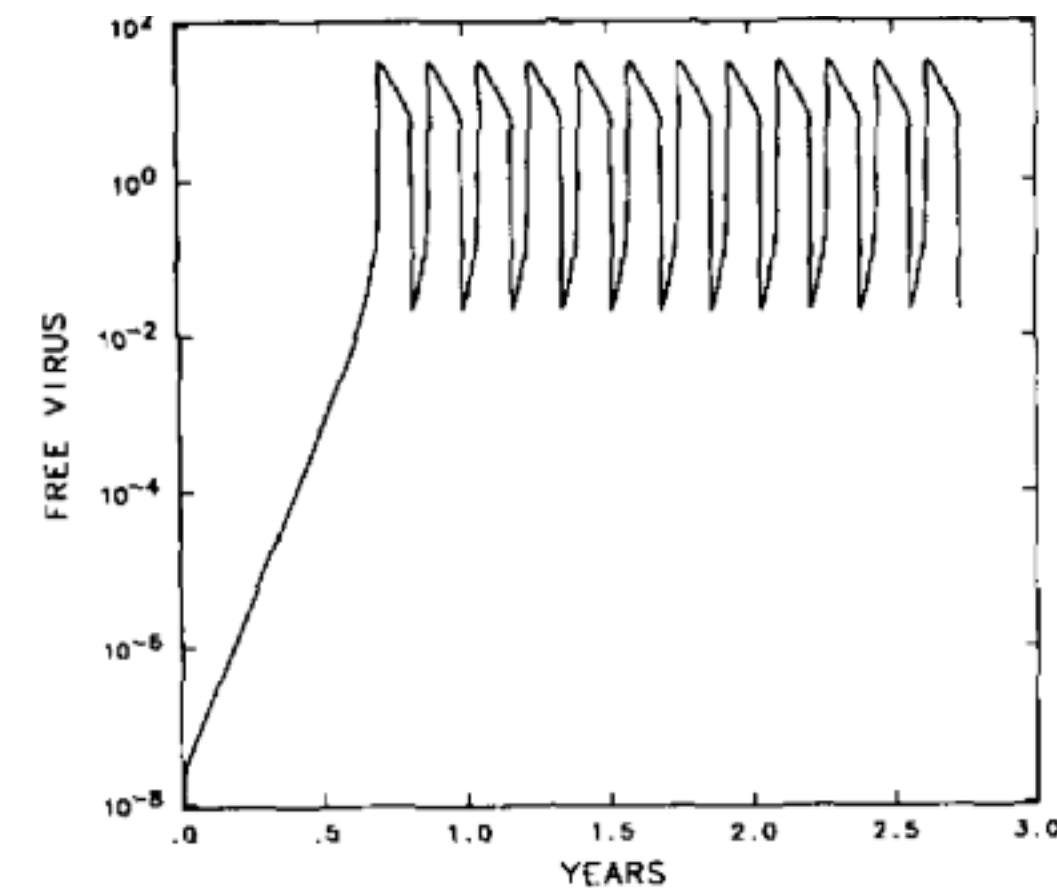
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$$\frac{dT}{dt} = s - \mu_T T + rT \left(1 - \frac{T + T^* + T^{**}}{T_{\max}}\right) - k_1 VT, \quad (5a)$$

$$\frac{dT^*}{dt} = k_1 VT - \mu_T T^* - k_2 T^*, \quad (5b)$$

$$\frac{dT^{**}}{dt} = k_2 T^* - \mu_b T^{**}, \quad (5c)$$

$$\frac{dV}{dt} = N\mu_b T^{**} - k_1 VT - \mu_V V. \quad (5d)$$



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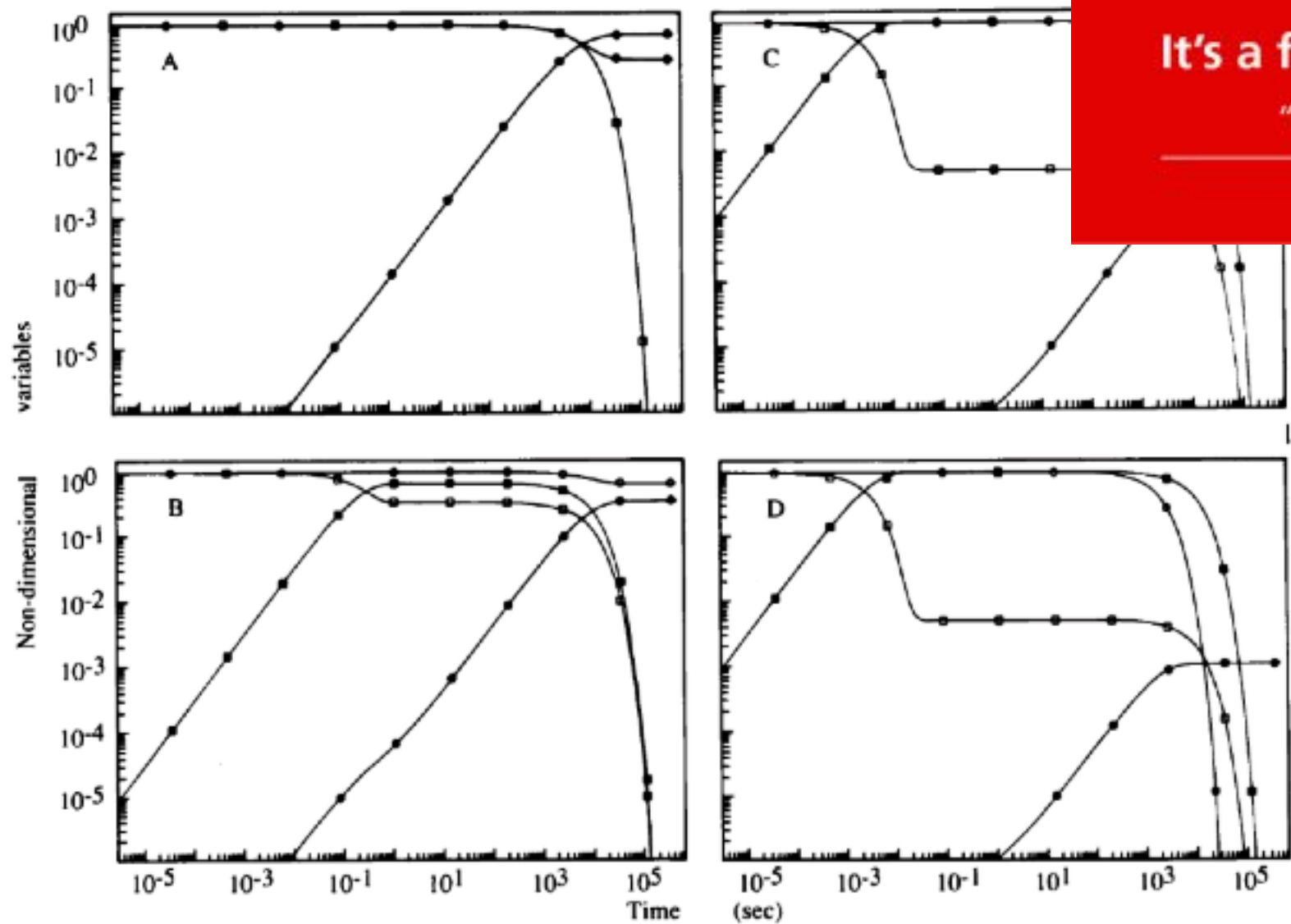
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SAVING
LIVES
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U = U

Undetectable = Untransmittable

It's a fact...

"a person living with HIV who has an undetectable viral load will not pass HIV to their sexual partners."

The Saving Lives charity is proud to support the Prevention Access Campaign's UequalsU consensus statement

$$\frac{dT^{**}}{dt} = \left(\frac{\lambda}{k_1} \right) - k_1 VT, \quad (5a)$$

(5b)

(5c)

(5d)

